

Encrypted Voice Calls with IP enabled Wireless Phones over GSM/ CDMA/ WiFi Networks

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Abstract—A research on the related technology and implementation of IP phone based on WiFi network is discussed in this paper; it includes the net structure of the technology used in designing the terminal of IP phone. This technology is a form of telecommunication that allows data and voice transmissions to be sent across a wide range of interconnected networks. A WiFi enabled IP phone is used which is preinstalled with the Symbian Operating System and a software application is developed using J2ME which allows free and secured communication between selected IP phones in the WiFi network. This communication is done with the use of routing tables organized in the WiFi routers. Using the free bandwidth of 2.4 GHz communication channels are established. The communication channel being a free bandwidth is vulnerable to external attacks and hacking. Thereby this challenge of creating a secure communication channel is addressed by using two different encryption mechanisms. The payload and header of the voice data packets are encrypted using two different algorithm techniques. Hence the communication system is made almost fully secure. Also the WiFi server can tunnel the calls to the GSM network using UNC. Moreover the creation of a vertical handoff protocol will provide seamless communication for users. This handoff is defined in the context of a diverse network environment. It is cost effective, it allows easier communication, is great for international usages, and it can be very useful for large corporations. In time this will become a cheap and secure way to communicate and will have a large effect on university, business and personal communication.

Keywords—IP phones, Wifi ,Symbian, J2ME, encryption, GSM, UNC.

I. INTRODUCTION

As human started to get civilized, great need for more, advance equipments occurs. Most of the things which in their early phase are considered to be part of leisure, have become one of the most necessary things of daily life. One such invention is telephone. However the development of conventional telephony systems is far behind the development of today's Internet. Centralized architectures with dumb terminals make exchange of data very complex, but provide very limited functions. Closed and hardware proprietary systems hinder the enterprise in choosing products from different vendors and deploying a voice function to meet their business needs. Consequently, Web-

like IP phone distributed architecture [1] is proposed to facilitate enterprises and individuals to provide their own phone services. The advent of Voice over Internet Protocol (VoIP) has fundamentally been transforming the way telecommunication evolves [2]. This technology is a form of telecommunication that allows data and voice transmissions to be sent across a wide variety of networks. VoIP allows businesses to talk to other branches, using a PC phone, over corporate Intranets. Driven by the ongoing deployment of broadband infrastructure and the increasing demand of telecommunication service, VoIP technologies and applications have led to the development of economical IP phone equipment for ever-rising VoIP communication market [3] based on embedded systems, IP phone application can satisfyingly provide the necessary interfaces between telephony signals and IP networks [4]. Although IP phone communication over the data networks such as LAN exists but these IP phones are fixed type. We have tried to implement wireless IP phone communication using the WiFi network. This network being in the free bandwidth channel is considered insecure and vulnerable to security threats and hacking. So the area of concern is the security and running cost of a communication system. As a lot of sensitive information can be lost because of insecure communication system, a lot of work is required to be done in this field to fill the lacuna. The base idea is unifying voice and data onto a single network infrastructure by digitizing the voice signals, convert them into IP packets and send them through an IP network together with the data information, instead of using a separate telephony network. Also the success of wireless and mobile communications has resulted in the creation of a large variety of technologies. Expanding services through the use and coordination of these diverse networks will provide user mobility between different types of systems. To achieve seamless mobility, network management operations must be conducted without causing degradation of services, and without need for user intervention. The movement of a user within or among different types of network can be referred to as intersystem or vertical mobility. One of the major challenges for seamless vertical mobility is vertical handoff, where handoff or handover is the process of maintaining a mobile user's

active connections as it changes its point of attachment. An issue in vertical handoff is the timely and reliable transfer of a mobile user's connections [11].

II. RELATED WORKS

The primary feature of a voice application is that it is extremely delay-sensitive rather than error-sensitive. There are several approaches that have been developed to support delay-sensitive applications on IP networks. In the transport layer, UDP can be used to carry voice packets while TCP may be used to transfer control signals, as long delay is caused by TCP by its retransmission and three-handshake mechanism. The Real-Time transport protocol (RTP) [5] is a compensative protocol for real-time deficiency on packet networks by operating on UDP and providing mechanisms for realtime applications to process voice packets. The Real-Time Control protocol (RTCP)[3] provides quality feedback for the quality improvement and management of the real-time network. Several signaling protocols have been proposed for IP phone applications. SIP is peer to-peer protocols. Being simple and similar to HTTP, SIP [6] will bring the benefits of WWW architecture into IP telephony and readily run wherever HTTP runs. It is a gradual evolution from existing circuit-switched networks to IP packet-switched network. A lot of work has been done to implement IP phones over data networks, even on the internet (Skype), but almost all work has been done mainly using secure communication channels and fixed IP phones. A lot of work has been done to connect different heterogeneous networks like UMA (Unified Mobile Access) technology that allows the use of both GSM network and WiFi networks (indoors) for calling [7].

III. EAS IP PHONE COMMUNICATION OVER WIFI

Fir IP based phone communication in a particular WiFi network is free. Moreover the communication is secured as the existing WiFi network is used rather than using the services of any other carrier. 128 bit encrypted voice communication takes place between authorized and authenticated IP phone users. If the user wants to call to outside world then he has suffix a symbol, in this case ' * '. Then the call is routed to the outside world. Also, if the user moves out of the WiFi range, handover takes place the mobile unit again starts working on GSM network.

A. Architecture

IP enabled cell phones are the mobile units capable of accessing the WiFi network. WiFi Routers have routing tables which are used to route the calls to the desired IP phone. A J2ME application was developed which provide access to the IP phone in the WiFi network.

B. Connection Mechanism

- IP phones registers its fixed IP on the WiFi route, where the router will update its routing table with this IP phone being active (Figure 1).

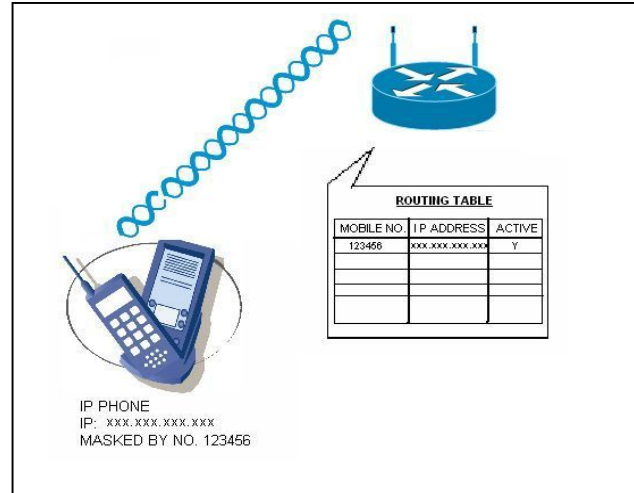


Figure 1: Registering of IP phone in the Routing table

- The name and number of the phone with the particular IP are searched in the database and IP is replaced with the name of the user in the WiFi routing table.
- If the number starts with a special symbol say asterisk ' * ' then the router tunnels the call to GSM network using UNC.
- When signal of WiFi fades out handover takes place and the mobile unit starts working on the GSM network.

C. Management of call between WiFi to WiFi

A number (user 2) is dialed using the J2ME application from user 1's mobile unit. The application then sends the number in 128 bit encrypted form to the router, requesting a call to be placed (Figure 2(a)). The router in the WiFi network searches its routing table for the desired number and if the number is active then a packet of data signaling an incoming call is sent to the corresponding IP on the WiFi network. The J2ME application on user 2's mobile unit alert's the user of an incoming call. The routing table gets updated to both the IP mobile units as being busy (Figure 2(b)). When the user 2 accepts the incoming call, real time transfer of voice data packets starts between the two mobile units. The header of each packet is encrypted in such a way that the router can decrypt it and route it to the required mobile unit. While the actual voice data packets are encrypted in such a way that only the other mobile unit can decrypt and it can not be decrypted at the router end. When the call is broken down the routing table is again modified and the busy status is changed. If the user at the other end doesn't want to take the call and presses the hang up button then the user at the first end is send a message that the user dialed is busy.

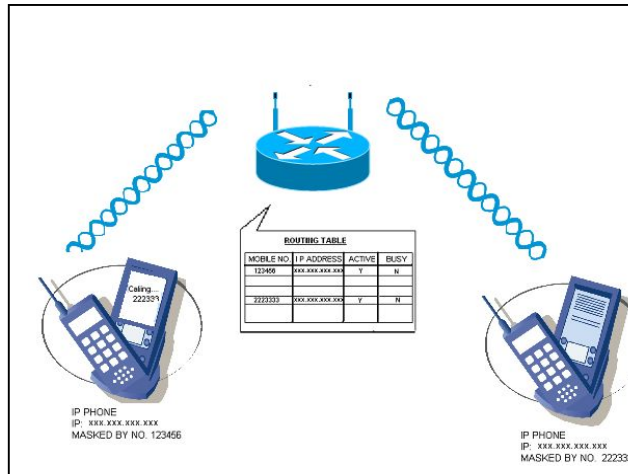


Figure 2(a): User 1 dialing User 2's number

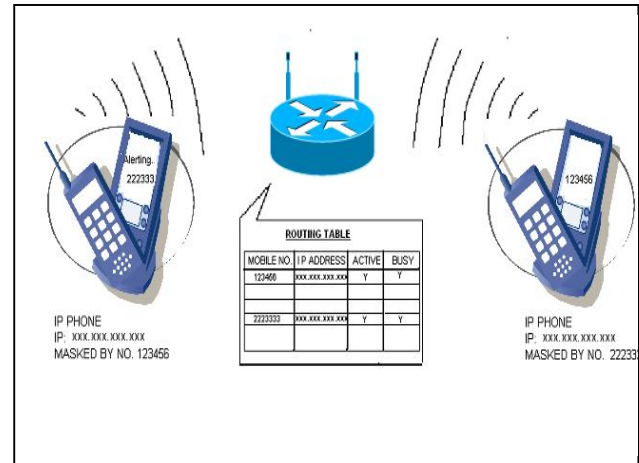


Figure 2(b): User 2 receiving a call from User 1

D. Management of call between WiFi to Public Network

When a user wants to dial a call to the outside world(that is to the public network) , he has to suffix an ' * ' before the number he wants to call to. If he dials "*1234567890",Then the WiFi router identifies the ' * ' and routes the call via broad band connection to the UNC (UMA Network controller). Till UNC IP was being used to carry the voice – data packets (Refer to Figure 3). After that point it depends upon the UNC which technology is used to carry the packets. Also if the call has to be routed to the outside world the packets have to be decrypted as the UNC is unaware of the encryption used by the WiFi network. Moreover the packet has to be organized and decrypted according to the needs of UNC.

E. WiFi to GSM handover

In case of WiFi to GSM handover, first the mobile unit has to detect that the WiFi signal has completely faded out. Also now the WiFi service is no longer acceptable. At this stage the mobile unit sends a handover request to a neighboring GSM cell. The selection of mobile cell depends upon the SIM card present in the mobile unit at that time. Then the core network of the service provider has to handle the resource allocation procedure with the base station controller (BSC) for the GSM calls. Once the allocation is complete a signal is sent to the mobile unit that the handover has taken place

F. Implementation

- A cell phone with Symbian 60 ver. 9 operating system ,with Java capabilities, also it should be equipped with WLAN 802.11 b/g capabilities.
- J2ME software is required to place the calls and allow the encryption to take place for a secured communication.

- A router with the routing tables is required to route the calls to specific online users. The router should be authenticated in the WiFi environment and should also be WLAN 802.11 b/g supported.

D. SECURITY

Security is one of the main areas of concern especially if we are communicating over the free bandwidth of 2.4 GHz of the WiFi .This is taken into care by using two different encryption methods. One is used for encrypting the header of the data packets this can be broken down by both the WiFi router and the mobile unit. While the payload is encrypted using a different method which can only be broken down at the other mobile unit(Refer to Figure 3). There are very less chances of the signals being tapped as this whole communication system is taking place on a private network, with authentic and limited connectivity. Also the area of coverage being limited the signals can not be tapped easily. It also handles the new upcoming threats which the employer is facing specially in defense and other sensitive organizations related to security and privacy of the organization because of the highly sophisticated mobile devices capable of audio and video recording .This communication system solves this problem by giving the employees mobile units which are Java enabled and capable of accessing the WiFi network, when the enter the organization and confiscating the employees mobile phones.

G. COST EFFICIENT

Cost involved in the setup and running of a communication system is a major issue. This method of communication deals very effectively in this aspect. As the only major cost involved is mainly in the setup of the communication system, which also comes out to be very less than the conventional GSM and CDMA networks. The running cost of the network is only of the calls routed through UNC to

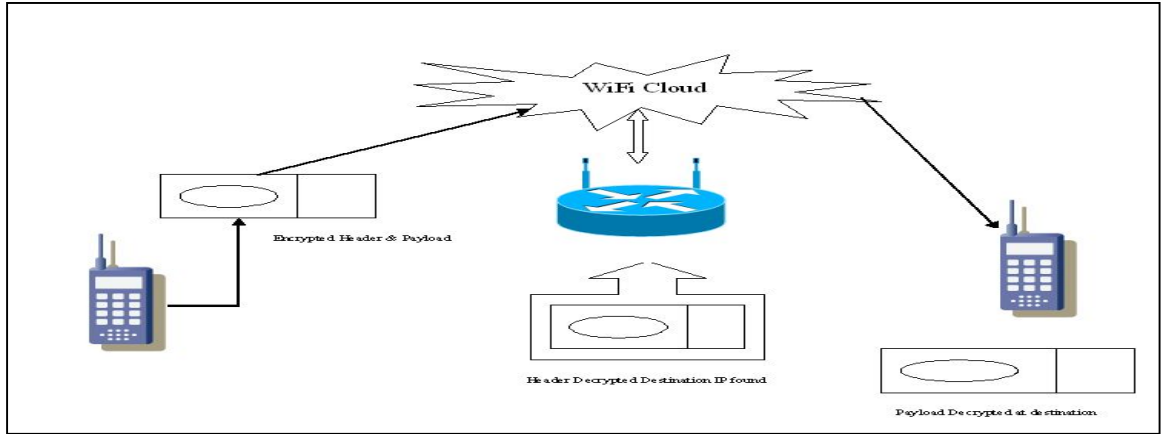


Figure 3: Encryption –Decryption mechanism of the channel

THEGSM network, which is the cost levied by the service provider while the calls made within the WiFi network are free of cost. Hence the running cost can be assumed to be nil as compared to the running cost of GSM and CDMA networks. Hence this communication system is very cost effective and cost efficient system.

H. COVERAGE

The coverage area of the network depends upon the WiFi router coverage. Unlike GSM network we can not deploy a number of WiFi hotspots for increasing the coverage. It is mainly due to the problem faced in handover. Any handoff operation is a three-stage process that includes handoff decision, radio link transfer and channel assignment [12]. The mobile unit will not attempt a handoff until the quality of signal deteriorates quite considerably. The base station usually measures the quality of the radio link channels being used by mobile nodes in its service area. This is done periodically so that degradations in signal strength below a prescribed threshold can be detected and handoff to another radio channel or cell can be initiated. While performing handoff, the mobile node's connection may be created at the target base station before the old base station connection is released. This is referred to as a make before break handoff. On the other hand, the new connection may be set up after the old connection has been torn down, which is referred to as a break before make handoff, which means that the mobile node can only communicate on a channel with one base station at a time. Furthermore, in some cases even if the handover break is short, the perceived voice quality can be very poor for several seconds due to low signal quality prior to the handover. To address the handover time several methods have been developed but in practice are not widely implemented nor supported by current devices [7][8][9][10][11].

I. FUTURE PROSPECTS

Through this paper we have tried to establish a new way of communication between two wireless IP phones over the WiFi network. However there are many areas which remain untouched and demand attention. There is a high potential for the development of applications for this communication system which in turn will transform this system into a full-fledged communication system. Applications like Short Messaging Service (SMS) can also be developed. This service will function between two IP phones on the same WiFi network or even a series of interconnected networks. Data exchange i.e. sharing and transfer of information and files between two IP phones is another application waiting to be developed. Again this service can function on the same WiFi network or a series of interconnected networks. Accessing and surfing the internet on the wireless IP phone through a single access point will be very cost efficient. Moreover acquiring a communication channel between the mobile unit and the router. The same bandwidth is used as a communication channel between the different WiFi networks thereby treating list of all the users that are logged on the network a real time chatting application can be developed. Moreover the interconnected IP phones can be linked to server like the Asterisk and dialing outside their native network to the outside world will be possible. This will be quite preferable as only a single line outside the network is needed which will allow access to all the connected IP phones.

IV. CONCLUSION

In this paper we have described a new way to provide communication within a specified area. Here we have proposed to use IP enabled mobile units which will be able to communicate to each other via the WiFi network. With the help of a simple Java application the allowed IP phones can automatically log on in the network and can communicate among themselves. The WiFi bandwidth of 2.4 GHz acts as

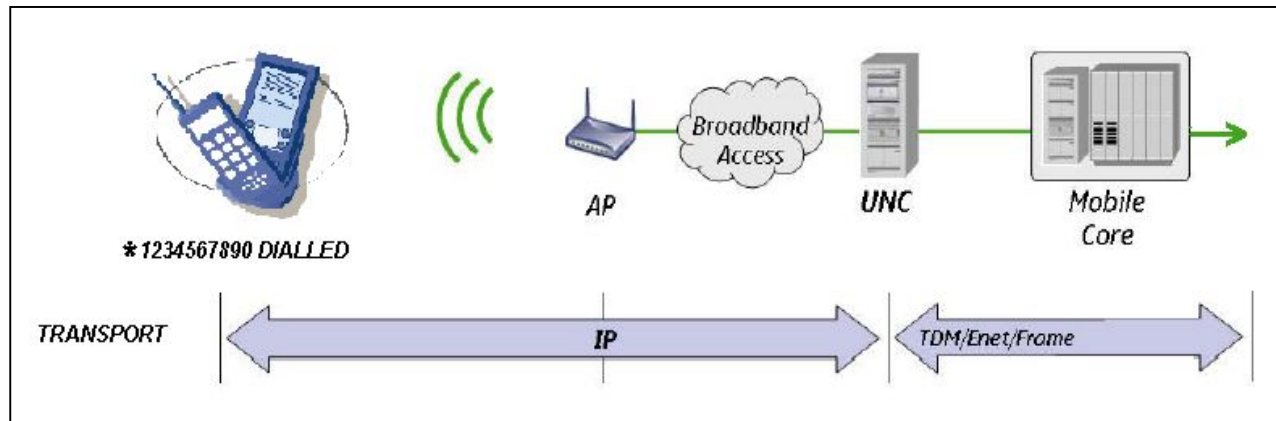


Figure 4: WiFi Network access to Terrestrial and Cellular Network

the whole network as one and creating a huge data cloud. Since the bandwidth of the WiFi network is free, the only cost involved in this communication system is the initial setup cost, hence making it very much viable. Although it limits the communication area but also provides the flexibility to dial calls to the outside world by tunneling the calls through UNC to the public network (terrestrial, GSM and CDMA network). At the same time it also addresses the security issues and is an answer to the no mobile zones. These are basically the zones where the organizations have prohibited the use of mobile phones because of certain security constraints such as the fear of leakage of sensitive information outside a desired area. This security in the communication channel is maintained as the data packets are 128 bit encrypted.

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